



11103 - Echoes Through the Intergalactic Medium: Building the Cosmic Noon FRB

Host Sample with JWST

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	pre-imaging	NIRCam Imaging	(1) FRB_Host_Generic

Folder	Observation	Label	Observing Template	Science Target
	2	spectroscopy of host	NIRSpec IFU Spectroscopy	(1) FRB_Host_Generic

ABSTRACT

Fast Radio Bursts (FRBs) are brief, luminous radio flashes that trace ionized gas across cosmological distances. Their dispersion measures encode the integrated electron column density along each sightline, directly linking FRB redshifts to the cosmic baryon distribution: the “Macquart relation.” Until recently, however, this relationship has been confirmed only at relatively low redshifts ($z < 1.3$), limited by the difficulty in detecting distant FRB host galaxies with ground-based facilities.

We propose JWST Target-of-Opportunity observations to identify and characterize host galaxies of high dispersion measure FRBs that will be discovered by the MeerTRAP and CRAFT surveys. Each triggered event will use deep NIRCам imaging in the F200W and F322W2 filters to pinpoint candidate host galaxies down to faint magnitudes ($m \sim 29$ AB). Subsequent NIRSpec integral-field spectroscopy (CLEAR/PRISM) will provide definitive redshifts and nebular emission-line diagnostics of the host. These observations will create the first sample of FRB host galaxies at $1 < z < 3$, spanning a critical epoch in cosmic history. This dataset will allow us to test whether FRBs effectively trace star formation, galactic feedback, and the distribution of ionized baryons in the intergalactic medium. JWST’s unrivaled infrared sensitivity and high spatial resolution are uniquely capable of detecting and obtaining redshifts for these faint, distant hosts. By extending the Macquart relation into the era of peak star formation, this program will open a powerful new window onto the cosmic baryon census and the structure of the cosmic web.

OBSERVING DESCRIPTION

We propose JWST Target-of-Opportunity (ToO) observations to rapidly identify and confirm host galaxies of high-redshift Fast Radio Bursts (FRBs). Each triggered observation will follow a two-stage observing strategy:

- (1) NIRCам Imaging: Deep imaging in the F200W and F322W2 filters will be conducted first, to identify candidate host galaxies down to faint magnitudes (~ 29 AB). The selected filters maximize sensitivity and enable accurate photometric identification of galaxies near the FRB localization.
- (2) NIRSpec IFU Spectroscopy: Once candidate hosts are identified from imaging, we will perform follow-up spectroscopy using the NIRSpec integral field unit (IFU) with CLEAR/PRISM to obtain precise redshifts and nebular emission-line diagnostics. The IFU provides comprehensive spatial coverage, ensuring unambiguous association of FRBs with their host galaxies and robust redshift confirmation.

Given the faintness and compact nature of expected host galaxies, JWST’s unique infrared capabilities and high spatial resolution are critical to achieving our science goals. Each trigger will require flexibility in scheduling, as well as careful coordination between imaging and spectroscopic follow-up.

Proposal 11103 - Targets - Echoes Through the Intergalactic Medium: Building the Cosmic Noon FRB Host Sample with JWST

Generic Targets	#	Name	Criteria	Description
	(1)	FRB_Host_Generic	FRB	
	<i>Comments: This is the generic target for the FRB location.</i>			
	(2)	TA_Star_Generic	Star	
	<i>Comments: This is a TA star for WATA which will be detemined from NIRCcam imaging of the FRB field.</i>			

Proposal 11103 - Observation 1 - Echoes Through the Intergalactic Medium: Building the Cosmic Noon FRB Host Sample with JWST

Observation	Proposal 11103, Observation 1: pre-imaging Diagnostic Status: Warning Observing Template: NIRCам Imaging										Fri Mar 13 21:01:36 GMT 2026
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria	Description							
	(1)	FRB_Host_Generic	FRB	Comments: This is the generic target for the FRB location.							
Template	Module	Subarray			Target Placement						
	ALL	FULL			Module B center (small extended source)						
Dithers	#	Primary Dither Type		Primary Dithers	Subpixel Dither Type		Dither Size		Subpixel Positions		
	1	INTRAMODULEX		8	STANDARD				1		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F200W	F322W2	DEEP2	20	1	8	8	32811.569		
Special Requirements	Target Of Opportunity Response Time 14 Days Fiducial Point Override NRCBS_FULL										

Proposal 11103 - Observation 2 - Echoes Through the Intergalactic Medium: Building the Cosmic Noon FRB Host Sample with JWST

Observation	Proposal 11103, Observation 2: spectroscopy of host											Fri Mar 13 21:01:36 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Comments: This observation will perform NIRSpec IFU spectroscopy of the FRB host galaxy identified from the preceding NIRCам imaging. A minimum interval of two weeks between the NIRCам and NIRSpec observations is required to allow for data reduction and host identification using the PATH analysis.											
Diagnostics	(spectroscopy of host (Obs 2)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	FRB_Host_Generic	FRB									
	Comments: This is the generic target for the FRB location.											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 TA_Star_Generic	WATA	FULL	F140X	NRSRAPID	3	1	1	42.947		
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2	20	1	false	true	NONE	4	4	5893.912	
	2	PRISM/CLEAR	NRSIRS2	20	1	false	true	NONE	4	4	5893.912	
	3	PRISM/CLEAR	NRSIRS2	20	1	false	true	NONE	4	4	5893.912	
	4	PRISM/CLEAR	NRSIRS2	20	1	false	true	NONE	4	4	5893.912	

Special Requirements	Target Of Opportunity Response Time 14 Days
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